claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ad831ff520e3b6a7c.jsonl`
- SHA-256: `58f1fcf785bf0dd90d06c25486f590854f3d9c9e4d5a9c5ff3bc27d12be7e0ba`
- Source modified: `2026-05-31T04:22:05+00:00`
- Imported at: `2026-07-05T16:48:21+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:21:06.032Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On the olmOCR-Bench evaluation, olmOCR 2 scores 84.9 on tables and 83.7 on multi-column layouts, with the paper attributing its largest gains specifically to table parsing and multi-column layouts ? the exact document features muneem needs for dense, multi-column bank statements."

Source: https://arxiv.org/html/2510.19817v1 (primary)
Supporting quote: "largest improvements in math formula conversion, table parsing, and multi-column layouts"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:08.608Z)

I'll verify this claim by examining the primary source (the olmOCR 2 paper) and searching for any contradicting or qualifying evidence.

3. assistant (2026-05-31T04:21:11.633Z)

I'll fetch the primary source first to verify the specific numbers and the attribution quote.

4. user (2026-05-31T04:21:19.749Z)

olmOCR 2 Benchmark Scores Analysis

(1) olmOCR-Bench Scores by Category

From Table 3, olmOCR 2's final scores are:
- **Tables**: 84.9
- **Multi-column**: 83.7
- **ArXiv**: 83.0
- **Old scans math**: 82.3
- **Headers & footers**: 96.1
- **Long tiny text**: 81.9
- **Base**: 99.7
- **Overall**: 82.4 ± 1.1

(2) Does olmOCR 2 score 84.9 on tables and 83.7 on multi-column?

Yes, exactly correct. Table 3 confirms these precise values in the final "olmOCR 2" row.

(3) Quote about "Largest Improvements"

The exact language states: "with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions."

This appears in the abstract and matches your characterization.

(4) Overall olmOCR-Bench Score

82.4 ± 1.1

Table 3 presents the complete development breakdown showing incremental improvements from olmOCR's initial 68.2 score to the final olmOCR 2 result of 82.4, representing a +14.2 point overall improvement as noted in the introduction.

5. user (2026-05-31T04:21:21.813Z)

Web search results for query: "olmOCR 2 olmOCR-Bench tables multi-column score 84.9 83.7"

Links: [{"title":"olmOCR 2: Unit test rewards for document OCR | Ai2","url":"https://allenai.org/blog/olmocr-2"},{"title":"olmOCR 2: Unit Test Rewards for Document OCR | alphaXiv","url":"https://www.alphaxiv.org/overview/2510.19817v1"},{"title":"allenai/olmOCR-2-7B-1025-FP8 · Hugging Face","url":"https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"},{"title":"(PDF) olmOCR 2: Unit Test Rewards for Document OCR","url":"https://www.researchgate.net/publication/396790587_olmOCR_2_Unit_Test_Rewards_for_Document_OCR"},{"title":"olmOCR 2 Unit Test Rewards for Document OCR","url":"https://arxiv.org/html/2510.19817v1"},{"title":"GitHub - allenai/olmocr: Toolkit for linearizing PDFs for LLM datasets/training · GitHub","url":"https://github.com/allenai/olmocr"},{"title":"olmOCR 2: Unit Test Rewards for Document OCR","url":"https://arxiv.org/pdf/2510.19817"},{"title":"OlmOCR-Bench: OCR Evaluation Benchmark","url":"https://www.emergentmind.com/topics/olmocr-bench"},{"title":"olmocr/olmocr/bench at main · allenai/olmocr","url":"https://github.com/allenai/olmocr/tree/main/olmocr/bench"}]

Based on the search results, I found information about olmOCR 2 and its performance on olmOCR-Bench, including the specific scores you mentioned.

olmOCR 2 achieves 84.9% on tables (up from 72.9%) and 83.7% on multi-column layouts (up from 77.3%). On olmOCR-Bench, olmOCR 2 achieves an overall score of 82.4 points?a nearly +4-point improvement over the previous release.

olmOCR 2 achieves state-of-the-art performance for real-world OCR of English-language digitized print documents and scores 82.4 points on olmOCR-Bench, delivering substantial improvements where OCR often fails. olmOCR 2 is built on Qwen2.5-VL-7B and fine-tuned on olmOCR-mix-1025, a dataset of 270,000 PDF pages with diverse properties?academic papers, historical scans, legal documents, brochures, and more.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:24.821Z)

The primary source confirms the numbers. Let me verify the attribution quote directly and check the official Ai2 blog to ensure the WebFetch isn't hallucinating, and to scrutinize the "largest gains" claim more carefully.

7. user (2026-05-31T04:21:32.405Z)

olmOCR 2 Performance Report

(1) olmOCR-Bench Per-Category Scores:

Yes, the scores are confirmed:
- **Tables: 84.9%** (up from 72.9%)
- **Multi-column: 83.7%** (up from 77.3%)

(2) Largest Improvements Quote:

The blog states: "olmOCR 2 delivers substantial gains where OCR traditionally struggles: 82.3% on old math scans (up from 79.9%), 84.9% on tables (up from 72.9%), 83.7% on multi-column (up from 77.3%)."

The three improvement areas mentioned are math formulas, table parsing, and multi-column layouts.

(3) Previous Version Scores:

- **Tables:** 72.9% (previous) ? 84.9% (olmOCR 2)
- **Multi-column:** 77.3% (previous) ? 83.7% (olmOCR 2)

(4) Base Model:

"olmOCR 2 is built on Qwen2.5-VL-7B and fine-tuned on olmOCR-mix-1025, a dataset of 270,000 PDF pages."

8. user (2026-05-31T04:21:37.304Z)

Web search results for query: "olmOCR 2 benchmark criticism limitations English-only OCR accuracy concerns"

Links: [{"title":"[Literature Review] olmOCR 2: Unit Test Rewards for Document OCR","url":"https://www.themoonlight.io/en/review/olmocr-2-unit-test-rewards-for-document-ocr"},{"title":"olmOCR 2 Unit Test Rewards for Document OCR","url":"https://arxiv.org/html/2510.19817v1"},{"title":"(PDF) olmOCR 2: Unit Test Rewards for Document OCR","url":"https://www.researchgate.net/publication/396790587_olmOCR_2_Unit_Test_Rewards_for_Document_OCR"},{"title":"olmOCR 2: Unit test rewards for document OCR | Ai2","url":"https://allenai.org/blog/olmocr-2"},{"title":"OlmOCR-Bench: OCR Evaluation Benchmark","url":"https://www.emergentmind.com/topics/olmocr-bench"},{"title":"deepseek-ai/DeepSeek-OCR-2 · Compare to LightOnOCR-2-1B","url":"https://huggingface.co/deepseek-ai/DeepSeek-OCR-2/discussions/1"},{"title":"olmOCR 2: Unit Test Rewards for Document OCR","url":"https://arxiv.org/pdf/2510.19817"},{"title":"[2510.19817] olmOCR 2: Unit Test Rewards for Document OCR","url":"https://arxiv.org/abs/2510.19817"},{"title":"Papers Explained 480: olmOCR 2 | by Ritvik Rastogi | Medium","url":"https://ritvik19.medium.com/papers-explained-480-olmocr-2-33fbf89757eb"}]

Based on the search results, here's what I found regarding olmOCR 2 benchmark criticism, limitations, and English-only OCR accuracy concerns:

Benchmark Scope: English-Language Limitation

olmOCR 2 achieves state-of-the-art performance on olmOCR-Bench, an English-language OCR benchmark. Although the benchmark is predominantly English and Latin-script-based, occasional samples may exercise multilingual tokenization and layout handling. This represents a significant limitation in terms of multilingual support and generalization beyond English documents.

Benchmark Design Criticisms and Limitations

A major limitation of OlmOCR-bench is its requirement for strict string consistency, which is often impractical in real-world scenarios.

However, the benchmark designers argue their approach improves upon traditional methods. While popular OCR benchmarks use edit distance against ground truth, olmOCR-Bench uses binary unit tests for two key properties: floating document elements like tables or figures lack a definitive ground truth representation, and unit tests allow different-yet-equivalently-correct representations to yield similar scores, while edit distance often rewards/penalizes these cases differently.

Areas of Strong Performance

Despite the limitations, the largest performance gains are observed in math formula conversion, table parsing, and multi-column layouts.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:21:42.663Z)

The claim is now strongly corroborated across two independent sources (the arXiv primary source and the official Ai2 blog). Let me do one final adversarial check ? verify the exact attribution wording in the abstract verbatim and confirm there's no misreading of what "largest improvements compared to previous versions" means versus what the claim states.

10. user (2026-05-31T04:21:45.127Z)

Based on the abstract provided, here is the relevant quote regarding improvements:

"We show that RL training on these test cases results in state-of-the-art performance on olmOCR-Bench, our English-language OCR benchmark, with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions."

However, the abstract does not state an overall olmOCR-Bench score. It only indicates "state-of-the-art performance" without providing a specific numerical benchmark result.

11. assistant (2026-05-31T04:21:56.713Z)

I now have sufficient evidence to assess the claim. Let me evaluate each part against the checklist.

Claim components:
1. "olmOCR 2 scores 84.9 on tables and 83.7 on multi-column layouts" ? CONFIRMED by Table 3 in the arXiv primary source and the official Ai2 blog (84.9 tables, 83.7 multi-column).
2. "the paper attributing its largest gains specifically to table parsing and multi-column layouts" ? CONFIRMED verbatim from the abstract: "with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions."

Adversarial scrutiny ? where the claim slightly stretches but does not break:
- The abstract attributes the largest gains to *three* areas: math formula conversion, table parsing, AND multi-column layouts. The claim quotes "table parsing and multi-column layouts" and frames these as "the exact document features muneem needs." This is a fair, on-topic subset ? it

doesn't misrepresent the source; it just omits the math-formula item (irrelevant to bank statements). Not an overreach.
- "Largest improvements... compared to previous versions" is a *relative delta* claim (vs. olmOCR v1), not an absolute "best categories." Indeed, the absolute per-category scores show Base (99.7) and Headers/Footers (96.1) are *higher* than tables (84.9). But the claim correctly says "largest gains" (delta), matching the source's "improvements... compared to previous versions" (72.9?84.9 tables = +12.0; 77.3?83.7 multi-column = +6.4). So the claim's framing is accurate.
- One minor caveat: the *overall* olmOCR-Bench score (82.4) is English-language only, and the benchmark uses binary unit-test scoring rather than strict edit distance. These are limitations of generalizing to Indian bank statements, but they do not contradict the specific numeric claim being verified.

Source quality: Primary arXiv source plus official Ai2 blog ? both high quality, current (Oct 2025), and directly support every numeric and attribution element. No contradicting source found.

The claim is well-supported, current, and the numbers match exactly. It does not refute.

12. user (2026-05-31T04:22:02.851Z)

Structured output provided successfully

13. assistant (2026-05-31T04:22:05.554Z)

Verification complete. The claim survives adversarial scrutiny: both numeric values (84.9 tables, 83.7 multi-column) and the attribution quote are confirmed verbatim in the primary arXiv source and corroborated by the official Ai2 blog. refuted=false, high confidence.